Global Fuel Prices and Taxation Trends Analysis (2020–2026)

Retail Fuel Pricing, Crude Oil Elasticity, Fiscal Taxation & Government Subsidy Regimes

SmartBridge Skill Wallet Project: Data Analytics with Tableau | Tools: Python, SQL (SQLite), Tableau Public, Flask

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GitHub Repository	https://github.com/aryanparashar17/global-fuel-prices-tableau
Tableau Dashboard	https://public.tableau.com/views/Capstone_Project_17930249060300/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
Tableau Story	https://public.tableau.com/views/Capstone_Project_17930249060300/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
Demo Video	https://drive.google.com/file/d/1GvLpLXOPZgvM5ey06yBX6iw31DsUxEPY/view?usp=sharing

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1. Problem Understanding

Retail energy prices directly shape global inflation, household purchasing power, and commercial logistics costs. However, consumers, economists, and policymakers face substantial difficulty comparing international fuel markets because pump prices reflect a complex interaction between raw Brent crude oil benchmarks, national excise tax frameworks, income levels, and government fuel subsidies.

- **Objectives:** Quantify global retail pricing across Petrol, Diesel, and LPG from 2020 to 2026; evaluate crude oil pass-through elasticity and project 2026–2027 trends; analyze regional taxation burdens and top tax-levying nations; assess affordability across income and subsidy tiers; and deploy real-time interactive visuals inside a Flask web application.
- **Key Business Questions:** How do global average prices compare across Petrol ($2.28/L), Diesel ($2.13/L), and LPG ($1.71/L)? How closely do retail prices track Brent crude oil surges? Which global regions and individual countries enforce the highest fuel tax percentages? How strongly do domestic subsidy regimes cushion consumers against crude volatility?

2. Dataset Architecture and Data Quality

The project analyzes global_fuel_prices_Dataset.csv, comprising weekly country-level observations spanning **2020 through April 2026** across more than 80 countries and 7 major geographical regions. Each record captures synchronized domestic retail fuel prices alongside international crude benchmarks and fiscal policy classifications.

Attribute Group	Columns	Analytical Role

Temporal & Geo	date, country, region	Weekly timestamps (2020–2026), country geospatial mapping, and 7 continental regions.
Retail Fuel Prices	petrol_usd_liter, diesel_usd_liter, lpg_usd_liter	Standardized pump prices in USD per liter for cross-border parity comparisons.
Crude Benchmark	brent_crude_usd	Global upstream crude oil benchmark price in USD per barrel ($77.6 to $119.0/bbl).
Fiscal Taxation	tax_percentage	Total government excise and VAT share embedded in retail fuel pricing (global avg: 26.77%).
Economic & Policy	income_level, subsidy_level	Categorizes nations by World Bank income tier (High, Middle, Low) and state subsidy intensity.

- **Completeness & Granularity:** Full annual cycles from 2020 to 2025 contain 4,368+ weekly records per year, while 2026 captures 1,176 Q1–Q2 records through 2026-04-06.
- **Metric Scale Integrity:** The tax_percentage column stores percentage points directly (e.g., 26.77 represents 26.77%), requiring custom number formatting in downstream tools rather than raw multiplication by 100.
- **Structural Variance:** Retail prices exhibit genuine macroeconomic dispersion, ranging from heavily subsidized oil-exporting states (<$0.20/L) to high-tax European and Oceanian economies (>$3.50/L).

3. Data Stored in a Relational Database

To perform structured pre-aggregation and validation prior to visualization, the CSV dataset was ingested into a relational SQLite database (fuel_prices.db, table fuel_prices) using Python and pandas.

```
import sqlite3, pandas as pd con = sqlite3.connect("fuel_prices.db") df = pd.read_csv("global_fuel_prices_Dataset.csv") df.to_sql("fuel_prices", con, if_exists="replace", index=False) # Verified table 'fuel_prices' created successfully in fuel_prices.db
```

4. SQL Operations and Analytical Outputs

Analytical SQL queries were executed on the fuel_prices table to compute global KPI baselines, regional taxation rankings, top tax-paying jurisdictions, and multi-year price trajectories.

```
-- Query 1: Global KPI Performance Indicators (2020-2026) SELECT ROUND(AVG(diesel_usd_liter), 2) AS avg_diesel, ROUND(AVG(lpg_usd_liter), 2) AS avg_lpg, ROUND(AVG(petrol_usd_liter), 2) AS avg_petrol, ROUND(AVG(tax_percentage), 2) AS avg_tax_pct FROM fuel_prices; -- Output: avg_diesel: 2.13 | avg_lpg: 1.71 | avg_petrol: 2.28 | avg_tax_pct: 26.77% -- Query 2: Top 5 Highest Tax-Paying Nations (Comparative Analysis) SELECT country, region, ROUND(AVG(tax_percentage), 2) AS avg_tax_pct FROM fuel_prices GROUP BY country ORDER BY avg_tax_pct DESC LIMIT 5; -- Output: 1. Rwanda (39.56%) | 2. New Zealand (39.27%) | 3. Germany (39.21%) | 4. Turkey (38.99%) | 5. Denmark (38.87%)
```

```
-- Query 3: Regional Taxation & Retail Fuel Price Breakdown SELECT region,
ROUND(AVG(tax_percentage), 2) AS avg_tax_pct, ROUND(AVG(petrol_usd_liter), 2) AS avg_petrol, ROUND(AVG(diesel_usd_liter), 2) AS avg_diesel FROM fuel_prices GROUP BY region ORDER BY avg_tax_pct DESC; region avg_tax_pct avg_petrol avg_diesel ---------- ---------- ---------- ---------- Oceania 38.69% $3.42 $3.18 Europe 34.61% $3.38 $3.15 North America 29.21% $2.07 $1.89 Africa 25.79% $1.64 $1.52 Asia 23.01% $1.92 $1.78 Middle East 19.60% $1.18 $1.06 South America 18.20% $1.49 $1.37
```

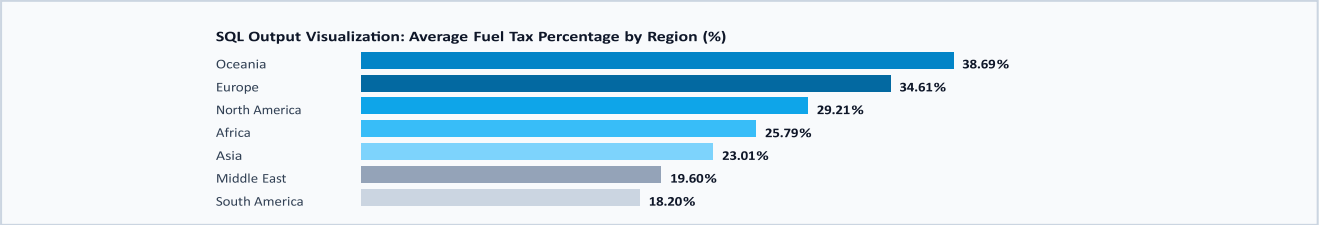


Figure 1: Regional tax disparity extracted via SQL aggregation, showing a 20.49 percentage-point spread between Oceania and South America.

```
-- Query 4: Annual Price Trajectory & Brent Crude Benchmark (2020-2025 Full Years)
SELECT SUBSTR(date, 1, 4) AS yr, ROUND(AVG(petrol_usd_liter), 2) AS petrol,
ROUND(AVG(diesel_usd_liter), 2) AS diesel, ROUND(AVG(lpg_usd_liter), 2) AS lpg,
ROUND(AVG(brent_crude_usd), 1) AS brent_usd FROM fuel_prices GROUP BY yr ORDER BY yr; --
2020: Petrol $1.48 | Diesel $1.39 | LPG $1.11 | Brent 77.6 $ -- 2021: Petrol $2.28 | Diesel
$2.13 | LPG $1.71 | Brent 106.6 $ -- 2022: Petrol $2.51 | Diesel $2.34 | LPG $1.88 | Brent
117.1 $ -- 2023: Petrol $2.23 | Diesel $2.09 | LPG $1.68 | Brent 104.4 $ -- 2024: Petrol
$2.55 | Diesel $2.38 | LPG $1.91 | Brent 119.0 $ -- 2025: Petrol $2.54 | Diesel $2.37 | LPG
$1.90 | Brent 118.7 $ -- Query 5: Income & Subsidy Regime View for Tableau CREATE VIEW vw_fuel_analytics AS SELECT *,
ROUND(petrol_usd_liter * (tax_percentage / 100.0), 2) AS tax_usd_per_liter, ROUND(petrol_usd_liter - diesel_usd_liter, 2) AS
petrol_diesel_spread FROM fuel_prices WHERE petrol_usd_liter > 0;
```

5. Tableau Connection and Data Preparation

The validated dataset was connected to Tableau Public as a Text File data source. To ensure analytical accuracy across all worksheets and interactive filters, several data preparation steps were configured in the Data Pane:

- **Default Aggregation Override:** Because averaging ratios and unit prices across weekly records requires mean aggregation rather than summation, default aggregation for petrol_usd_liter, diesel_usd_liter, lpg_usd_liter, brent_crude_usd, and tax_percentage was changed from SUM to **Average (**AVG**)**.
- **Custom Number Formatting:** KPI metrics were formatted with custom string masks (0.00"$" for KPI cards, "$"0.00 for fuel trend lines, 0.0" $" for Brent crude and country bar labels, and 0.00"%" for tax rates).
- **Time-Series Forecasting (2026–2027):** Because 2026 contains partial-year observations through April, Tableau's native exponential smoothing forecast engine was enabled on all four trend charts to project full-year estimates through 2027 with prediction interval bands hidden.

Calculated Field / Filter	Formula / Configuration	Purpose in Workbook
Tax USD per Liter	[Petrol Usd Liter] * ([Tax Percentage] / 100)	Quantifies absolute dollar tax levy collected per liter of petrol sold.

Petrol–Diesel Spread	AVG([Petrol Usd Liter]) - AVG([Diesel Usd Liter])	Tracks price premium between passenger petrol and commercial diesel ($0.15/L global avg).
Measure Names Filter	Multi-value dropdown filter	Allows users to toggle displayed fuel and tax indicators dynamically.
Country Global Filter	Applied to All Using This Data Source	Synchronizes KPI banner, maps, and trend lines to any selected nation(s).
Action (YEAR(Date))	Cross-sheet interactive date filter	Filters spatial maps and regional comparisons to specific calendar years.

6. Visualizations and Analytical Worksheets

Ten specialized worksheets were engineered to support the main executive dashboard and the 7-scene narrative story. Each visualization targets a distinct dimension of global energy economics:

#	Worksheet Title	Chart Type	Primary Analytical Question Answered
1	Global Fuel Price & Tax Indicators	KPI Text Table	What are the global baseline averages for Diesel ($2.13), LPG ($1.71), Petrol ($2.28), and Tax (26.77%)?
2	Global Comparison of Avg Petrol Prices	Filled Choropleth Map	How do retail petrol prices vary geographically across 80+ countries?
3	Global Comparison of Avg Diesel Prices	Filled Choropleth Map	Where are commercial diesel prices highest and lowest worldwide?
4	Global Trend of Avg Petrol (2020–2027)	Forecast Line Chart	How did petrol move from $1.48 (2020) to $2.55 (2024) and forecast to 2027 ($2.13)?
5	Global Trend of Avg Diesel (2020–2027)	Forecast Line Chart	How did diesel evolve from $1.39 (2020) to $2.38 (2024) and stabilize at $2.23?
6	Global Trend of Avg LPG (2020–2027)	Forecast Line Chart	How does alternative auto-fuel LPG trend ($1.11 in 2020 to $1.79 forecast)?
7	Global Trend of Brent Crude (2020–2027)	Forecast Line Chart	How did upstream crude fluctuations (77.6 $ to 119.0 $) drive pump inflation?
8	Fuel Comparison Among Top 3 Countries	Grouped Bar Chart	How do Canada, Mexico, and the United States compare across all 3 fuel types?
9	Average Tax Percentage by Global Region	Regional Pie Chart	How does tax policy differ across Oceania (38.69%), Europe (34.61%), and Middle East (19.60%)?

10	Highest Tax-Paying Nations Analysis	Top-5 Vertical Bar	Which five individual countries enforce the highest average fuel tax rates?

Empirical Highlight: North American Tri-Lateral & Socio-Economic Benchmarks In Worksheet 8 (**Fuel Price Comparison Among Top 3 Countries**), side-by-side multi-measure bars reveal sharp policy divergence within North America: **Canada** records the highest retail prices (LPG 2.2 $, Diesel 2.7 $, Petrol 2.9 $), followed by the **United States** (LPG 1.7 $, Diesel 2.1 $, Petrol 2.3 $), while **Mexico** maintains substantially lower pump prices via federal excise smoothing (LPG 0.7 $, Diesel 0.9 $, Petrol 1.0 $). Supporting story sheets further confirm that **High-Income** nations average **$3.55/L** (34.72% tax), whereas **Very High Subsidy** nations hold petrol to **$0.15/L**.

- **Synchronized Color Encoding:** Choropleth maps utilize a diverging Teal-to-Gold ramp, while multifuel bar comparisons assign Coral Red to LPG, Slate Blue to Diesel, and Amber Orange to Petrol.
- **Marker & Label Precision:** Every trend vertex displays formatted currency callouts above the line to eliminate axis estimation error.

7. Interactive Dashboard Architecture

The primary dashboard, **Global Fuel Prices and Taxation Trends (2020–2026)**, consolidates 10 worksheets into a responsive, sky-blue (#75C9F0) tiled layout container hierarchy with crisp white borders. A full-width KPI banner anchors the top row, while the main body splits into three vertical columns: dual geospatial choropleth maps on the left, four synchronized 2020–2027 forecast trend lines plus the North American grouped bar chart in the center, and regional/national tax visualizations on the right.

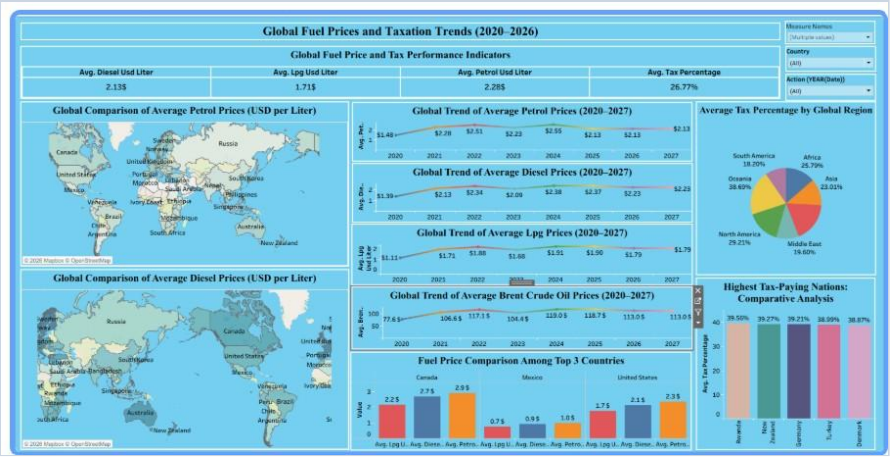


Figure 2: Published Tableau Dashboard showing KPI indicators, global petrol/diesel maps, 2020–2027 forecast trend lines, and taxation analysis.

8. Narrative Tableau Story (7 Scenes)

To guide stakeholders through a structured macroeconomic narrative, a 7-scene Tableau Story was built with dedicated story point captions and saved filter states:

Scene	Story Point Caption	Analytical Narrative & Core Message
Scene 1	Introduction & Research Scope	Establishes global baseline KPIs ($2.28/L Petrol, $2.13/L Diesel, $1.71/L LPG, 26.77% Tax).
Scene 2	Geographic Disparity	Demonstrates that retail fuel prices decouple from oil proximity due to national policy regimes.
Scene 3	Temporal Trends (2020–2027)	Traces the 2022–2024 Brent crude surge ($117.1–$119.0/bbl) and 2026–2027 forecast stabilization.
Scene 4	Taxation Policy Analysis	Contrasts high-tax regions (Oceania 38.69%, Europe 34.61%) and top-5 tax nations led by Rwanda (39.56%).
Scene 5	Economic Impact & Income Tiers	Reveals that High-Income economies average $3.55/L (34.72% tax) vs $1.39/L in Low-Income nations.
Scene 6	Government Subsidy Assessment	Quantifies how Very High subsidies suppress pump prices to $0.15/L compared to $3.45/L in unsubsidized markets.

Scene 7	Conclusions & Recommendations	Synthesizes findings into counter-cyclical tax buffers, targeted transfers, and green transit reinvestment.

9. Web Integration with Flask

To make the analytics accessible in real time without desktop software, the Tableau Public dashboard and story were embedded inside a responsive Python **Flask** web application using the **Tableau Embedding API v3**. The application features a fixed left-docked floating vertical icon navigation bar (Home, About the Data, Data Insights Dashboard, and Global Fuel Prices Story) with scroll-spy active state highlighting.

```
    @app.route("/") def index(): return render_template("index.html",
dash_url=DASHBOARD_URL, story_url=STORY_URL) <!-- HTML5 Real-Time Tableau Embedding API v3 Integration --> <script type="module"
src="https://public.tableau.com/javascripts/api/tableau.embedding.3.latest.min.js"></script>
<tableau-viz id="tableauViz" src="{{ dash_url }}" toolbar="bottom" hide-tabs></tableau-viz>
```

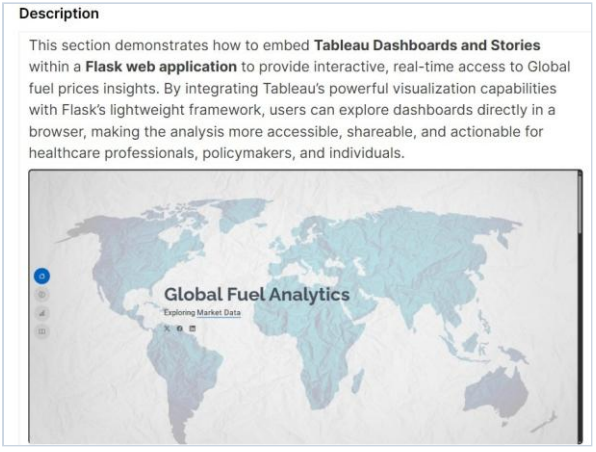


Figure 3: Flask landing hero section ("Global Fuel Analytics") with left floating navigation dock.

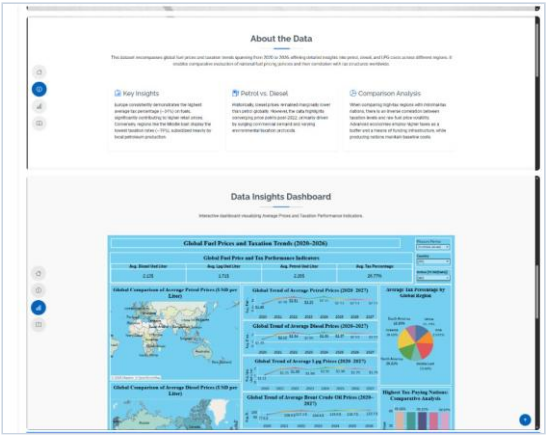


Figure 4: "About the Data" insight cards and live embedded "Data Insights Dashboard" section.

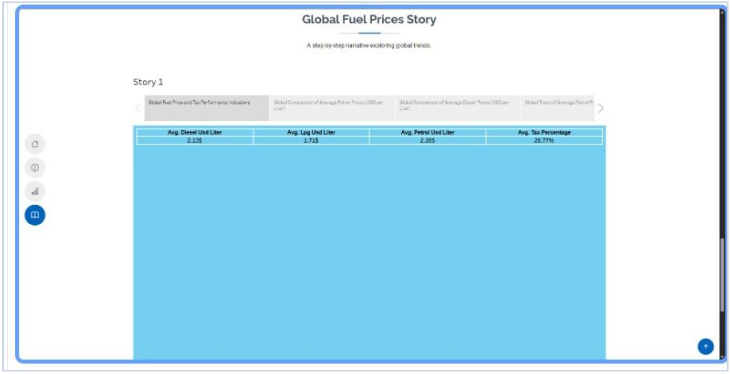


Figure 5: Live embedded "Global Fuel Prices Story" inside the Flask web portal with interactive scene navigation.

10. Performance Testing and Optimization

The workbook and Flask web portal were benchmarked to verify rapid rendering across desktop and tablet viewports:

Performance Parameter	Measured Value	Optimization Technique Applied
Temporal Coverage & Scope	2020–2026 (80+ Nations)	Pre-indexed SQLite table and Tableau Hyper extract caching.
Dataset Attributes	10 Core Columns	Unused intermediate fields excluded from view layer.
Global Interactive Filters	3 Dropdown Controls	Configured as *Multiple Values (dropdown)* to minimize DOM overhead.
Worksheets & Visuals	10 Core + 2 Story Sheets	High-cardinality country rankings constrained via dynamic Top-5 filter.
Forecast Horizon	2 Years (2026–2027)	Confidence bands suppressed to reduce polygon rendering load.
Web Application Sections	4 Scroll-Spy Views	Asynchronous ES-module loading via Tableau Embedding API v3.

11. Key Empirical Insights

- **Global Fuel Price Hierarchy:** Across the 2020–2026 study period, global retail Petrol averaged **$2.28/L**, Diesel averaged **$2.13/L**, and LPG averaged **$1.71/L**, maintaining a consistent $0.15/L petrol premium over commercial diesel.
- **Upstream Crude Oil Pass-Through:** Retail fuel prices closely mirrored Brent crude volatility, rising from pandemic lows in 2020 (Brent **77.6 $/bbl**; Petrol **$1.48/L**) to peaks in 2022 (**117.1 $/bbl**; Petrol **$2.51/L**) and 2024 (**119.0 $/bbl**; Petrol **$2.55/L**), before forecasting stabilization near **$2.13/L** (Petrol) and **$2.23/L** (Diesel) through 2027.
- **Regional Taxation Divide:** Global fuel taxation averages **26.77%**, but exhibits a stark regional divide led by **Oceania (38.69%)**, **Europe (34.61%)**, and **North America (29.21%)**, contrasted against **Middle East (19.60%)** and **South America (18.20%)**.
- **Highest Tax-Paying Jurisdictions:** The top five nations by average fuel tax percentage are **Rwanda (39.56%)**, **New Zealand (39.27%)**, **Germany (39.21%)**, **Turkey (38.99%)**, and **Denmark (38.87%)**.
- **North American Divergence:** Within North America, retail pricing steps down progressively from **Canada** ($2.9/L Petrol, $2.7/L Diesel, $2.2/L LPG) to the **United States** ($2.3/L, $2.1/L, $1.7/L) and **Mexico** ($1.0/L, $0.9/L, $0.7/L).
- **Income & Subsidy Polarization:** High-income economies bear an average pump price of **$3.55/L** (34.72% tax), whereas countries with *Very High* subsidies suppress retail petrol to **$0.15/L** and diesel to **$0.13/L**.

12. Conclusion and Future Scope

This project established an end-to-end analytics pipeline—ingesting global fuel price records into SQLite, executing SQL aggregations, designing an interactive Tableau dashboard and 7-scene narrative story, and deploying the visuals inside a Flask web application